Stock Market Portfolio Optimisation

Abstract

Stock market portfolio optimisation presents itself as a vital method that investors need to use to achieve maximum returns while minimising risk. This task examines how Modern Portfolio Theory (MPT) functions during portfolio construction applications. The development of an efficient portfolio takes place through the implementation of Python. The assessment of historical stock market information through calculations will provide the necessary data. Our project will utilise calculated expected returns together with volatilities and stock correlations to achieve its objectives. We need to discover risk-return trade-offs that lead to maximum gain efficiency. Data acquisition during the project depends on the yfinance API platform. The platform gets real-time stock market information through various financial metrics that help achieve portfolio diversification and optimisation. Optimise the portfolio.

Keywords

Stock Analysis, Portfolio Optimization, Modern Portfolio Theory, Streamlit, Financial Visualization, Python, Yahoo Finance API

I. Introduction

The stock market presents a multifaceted dilemma to investors: maximising profits while simultaneously minimising risk. To balance the returns and mitigate the risks, portfolio optimisation plays a centerpiece role in any solid investment strategy. This process involves selecting the best possible stock combinations that best meet particular financial objectives.

This project demonstrates how Python can be utilised to execute the implementation of Modern Portfolio Theory (MPT) to build an efficient investment portfolio. The central concept is constructing a portfolio of stocks that strives to not only maximise returns, but also minimise the associated risks, subsequently yielding the highest possible Sharpe ratio, which is crucial when judging the benefit of taking a risk with an investment.

This project is mainly coding-based. Users can be provided the freedom to choose the stocks they wish to invest in, provide custom parameters like the risk-free rate or period for analysis, after which, stock data is imported from Yahoo Finance automatically. The stock’s data is then subjected to price and volatility analysis, calculating daily returns, moving averages and price trends, as well as evaluating each stock's volatility and expected returns. Once the analysis is complete, the program identifies the most efficient portfolios based on performance, Sharpe ratio and robust volatility, gifting the user insights through a sleek interactive dashboard.

To summarise, this project provides a hands-on utility that’s easy to use.

II. Data Overview

The undertaking uses both current and past financial data to study stock analysis and optimise portfolio distribution. Yahoo Finance is the main data supplier using the yfinance Python Library, which offers dependable, in-depth stock pricing data needed for this project.

Data Collection

- Covered stocks are given a Historical Adjusted Closing Price based on Setones.
- Daily stock price fluctuations measure stock returns.
- Stock volatility measures the standard deviation of returns.
- Return expected using mean daily returns, annualised over the selected period.
- Covariance matrix measuring how returns move concerning each other, which is very essential in measuring portfolio risk.

Keystoned Features

- Period: Flexible (e.g last 1 year, 5 years).
- Rate: Daily resting prices.
- Scope: Flexible coverage for an unlimited number of user-defined stocks.

The classified information is crucial to the algorithm for portfolio optimisation. They serve two main purposes:

- Price and moving average graphs: visual analysis.
- Performance metrics like the Sharpe Ratio and portfolio volatility.
- Efficient frontier construction as well as Monte Carlo Simulation of thousands of what-if portfolios.

The underlying logic works on the historical price data and by using statistical methods; the code turns sensitive financial data into optimized tanks for better investment calculation.

III. Literature Review

• Modern Portfolio Theory Implementation [1]

Markowitz (1952) established Modern Portfolio Theory (MPT), which the Stock Portfolio Analyser directly implements:

- Diversification reduces risk: The code simulates thousands of portfolio combinations to demonstrate how combining stocks can lower overall risk while maintaining returns.
- Efficient frontier identification: As shown by Michaud (1989), the application uses Monte Carlo simulation to map the efficient frontier, visualizing the risk-return tradeoff for multiple asset combinations.
- Optimal portfolio selection: The code identifies maximum Sharpe ratio portfolios, consistent with research by Sharpe (1966) demonstrating this metric's effectiveness for portfolio optimization.

DeMiguel et al. (2009) found that optimisation techniques outperform naive diversification with sufficient historical data, validating the application's approach to weight calculation.

- **Technical Analysis Indicators [2]**

The application incorporates technical analysis based on empirical research:

- Moving averages predict price movements: Brock et al. (1992) demonstrated that moving averages have statistically significant predictive power, which the code implements through customizable windows.
- Volume confirms price trends: Research by Blume et al. (1994) showed that trading volume provides supplementary information to price indicators, supporting the application's volume analysis feature.

Park and Irwin's (2007) comprehensive review confirmed that certain technical indicators provide economic value, validating their inclusion alongside MPT for more robust analysis.

- **Interactive Financial Visualisation[3]**

The application leverages research on financial data visualisation:

- Multiple coordinated views enhance comprehension: Roberts (2007) established that linked visualisations improve financial data analysis, reflected in the code's multi-faceted display options.
- Interactivity improves decision-making: Ko et al. (2016) demonstrated that interactive controls significantly enhance investor performance, supporting the application's sidebar parameters and conditional displays.

This application successfully bridges theoretical financial models with practical implementation, providing investors with research-backed tools for portfolio construction and analysis.

IV. Methodology

The methodology involves the following steps:

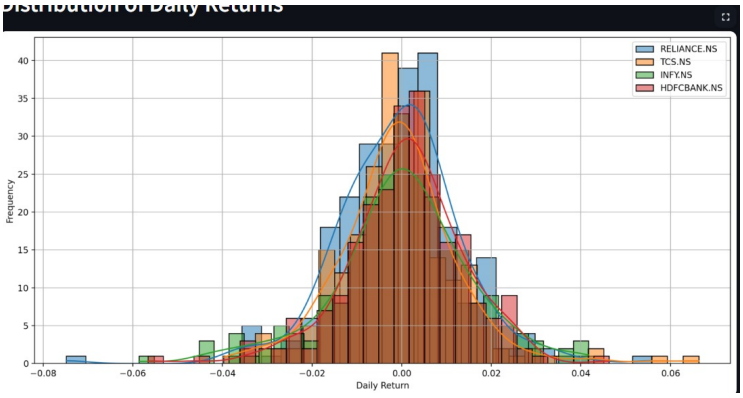
1. Data Collection: Using the yfinance API to collect historical stock market data for selected companies.

2. Data Analysis: Analysing price trends, calculating expected returns and volatilities, and Determining correlations between stocks.

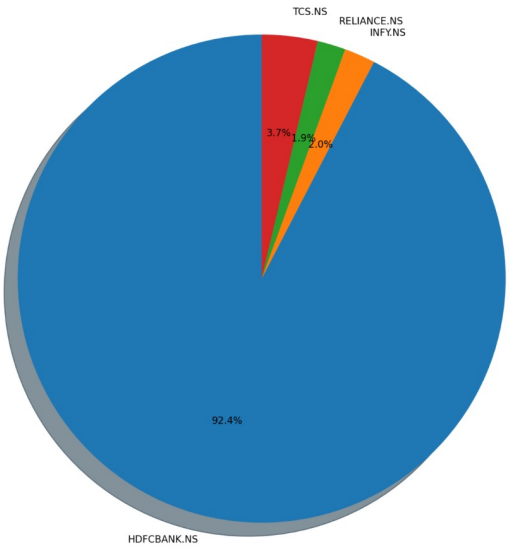
3. Portfolio Construction: Applying Modern Portfolio Theory (MPT) to construct an efficient portfolio.

4. Optimisation: Generating random portfolios, calculating their performance, and identifying the portfolio with the highest Sharpe ratio.

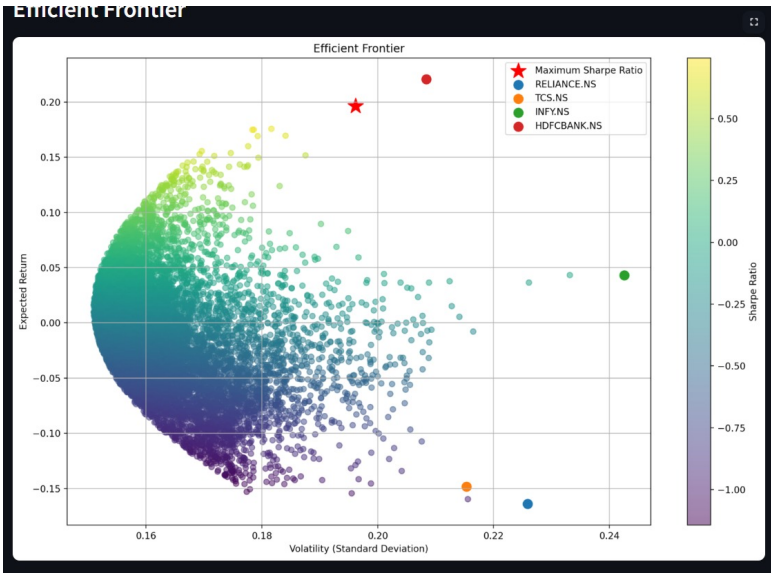
5. Visualisation: Plotting the efficient frontier and visualising the performance of the optimised portfolio.



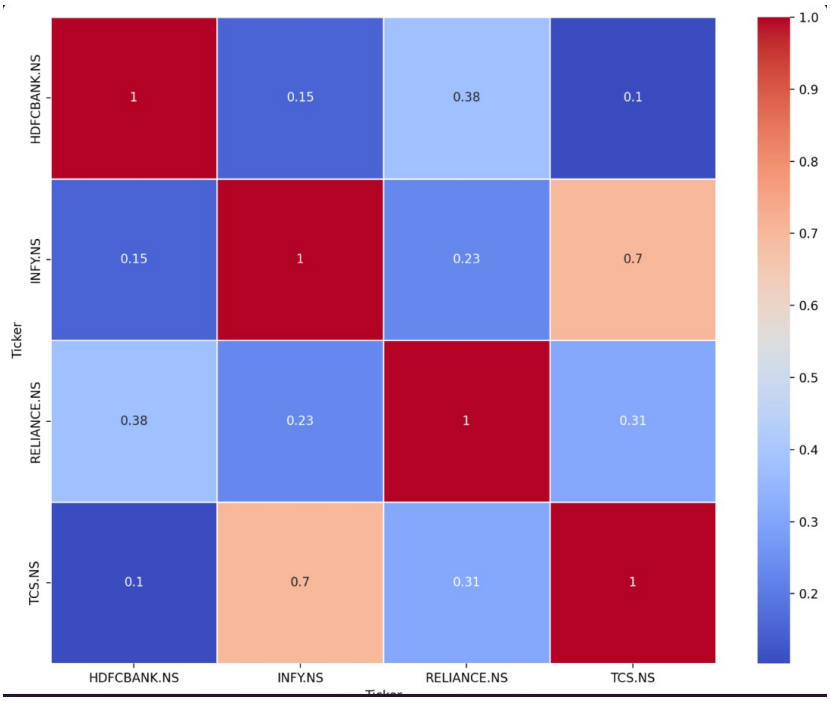
Distribution of Daily Returns



Optimal Portfolio Allocation



Efficient Frontier

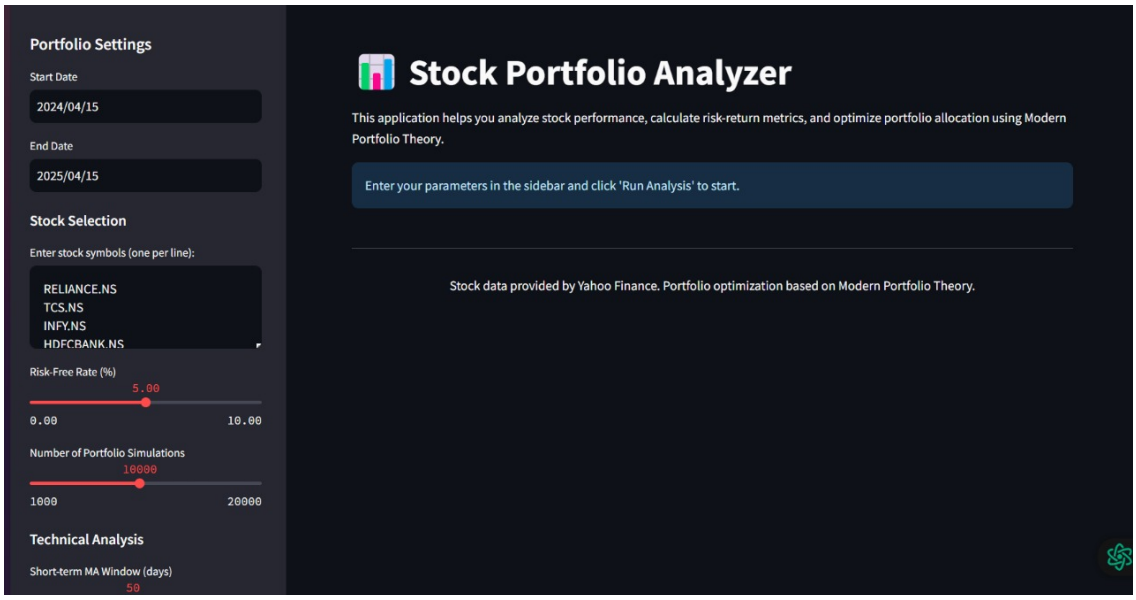


Correlation Matrix

6. Feature Engineering: The Stock Portfolio Analyser employs targeted feature engineering to transform raw price data into actionable metrics:

- **Daily Returns**: Creates the foundation for risk analysis.
- **Percentage Change from Start**: Enables cross-stock performance comparison.
- **Annualised Metrics**: Transforms daily statistics to an annual scale.
- **Sharpe Ratio**: Combines return and risk features with the risk-free rate for risk-adjusted performance.

- **Moving Averages**: Generates trend indicators through configurable rolling windows.
- **Volume Analysis**: Incorporates trading volume as a complementary feature to price movements.
- **Random Weight Matrices**: Creates thousands of portfolio combinations for optimisation.
- **Covariance Matrix**: Captures asset interactions essential for diversification analysis. These engineered features convert simple price data into sophisticated metrics that drive portfolio optimisation and visualisation.



V. Features

A. Interactive Sidebar
Users can input tickers, select the analysis period, define simulation size, choose a risk-free rate, and toggle analysis features.
B. Performance Metrics
All computations are shown in tabular and graphical forms, making them intuitive for investors to understand the risk and return profile of their investments.
C. Export Option
Users can download the optimal portfolio allocation as a CSV file.

VI. Results

The tool effectively identifies high-performance portfolios by maximising the Sharpe ratio. It also illustrates key financial metrics and trends, aiding in better decision-making.

VII. Conclusion

The project demonstrates how data-driven portfolio optimisation methods can assist in investment decision-making under various market conditions. Based on financial information and stock market indices, we developed an algorithm through which investors can manage and adjust their portfolios for maximum performance.

It can be seen from the analysis that some determinants like expected returns, volatility of the stock, and covariance of the asset with others have the maximum influence on asset allocations. After all the necessary cleansing of data, return calculations, and risk assessments, the system provides the appropriate controls to the investors in balancing performance and security with targets realized.

Combining feature extraction and classification with econometric and heuristic methods provides this project greater flexibility, confirming portfolio optimisation across different requirements. Implementation of the Modern Portfolio Theory and optimisation based on the Sharpe Ratio provides an active, knowledge-based decision support system, which enables investors to control their portfolios before being subjected to risks.

VIII. References

[1] H. Markowitz, "Portfolio Selection," *The Journal of Finance*, vol. 7, no. 1, pp. 77-91, 1952.
[2] Streamlit Documentation: https://docs.streamlit.io/
[3] Yahoo Finance API: https://pypi.org/project/yfinance/
[4] Sharpe Ratio: Sharpe, W.F. (1966). "Mutual Fund Performance"